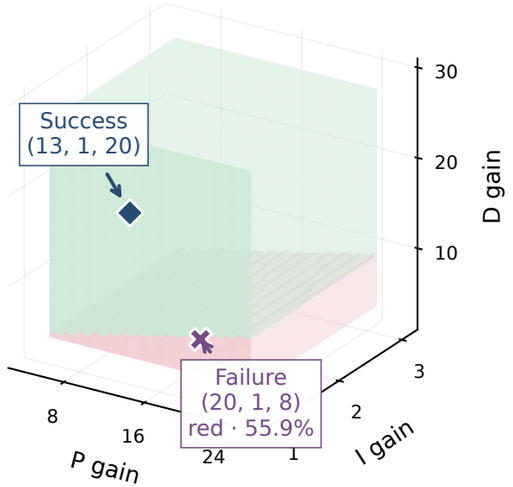
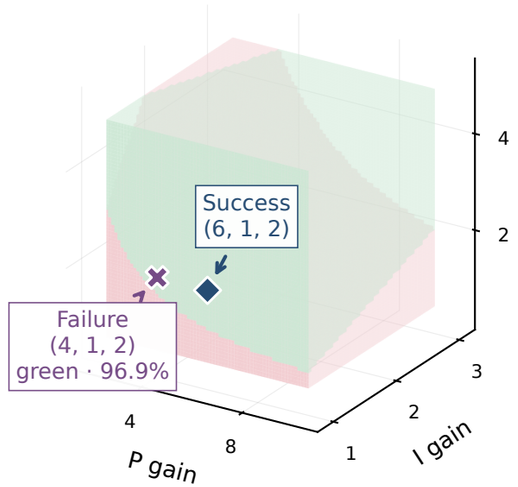
All PID gains: Success is in the green region on every axis

Failure 1

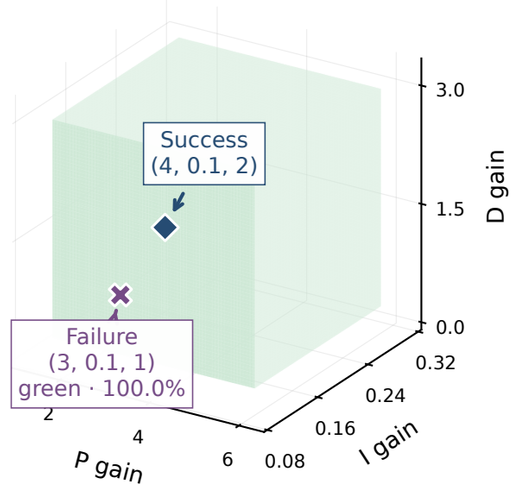
Roll / Pitch
Success: green (100%)



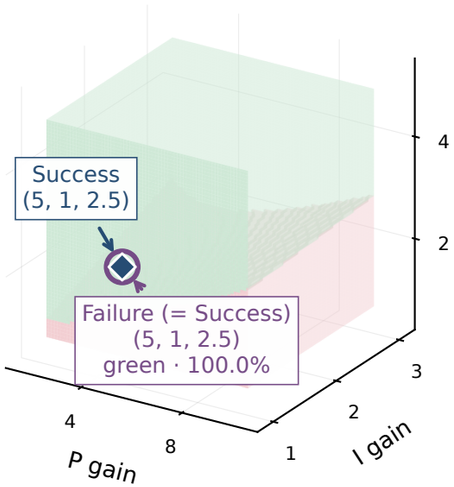
Yaw
Success: green (100%)



XY
Success: green (100%)

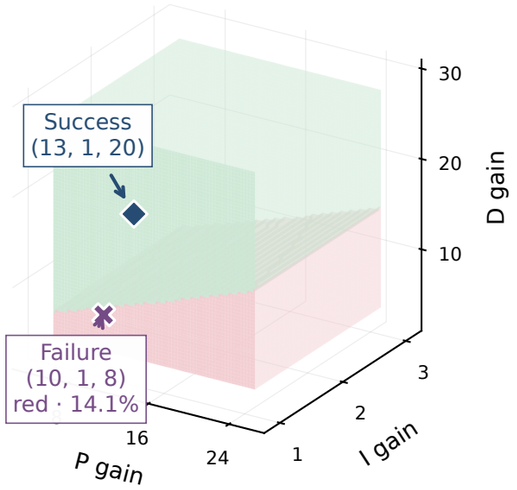


Z
Success: green (100%)

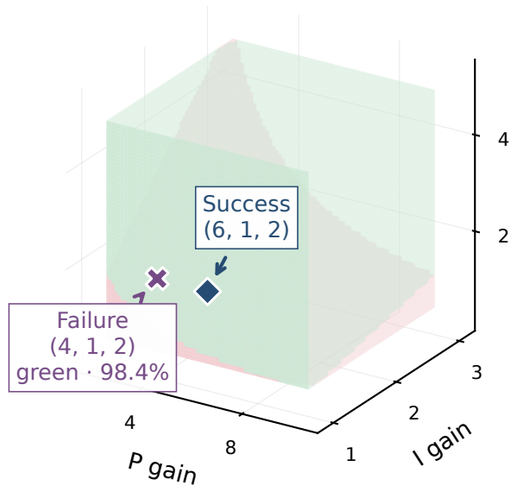


Failure 2

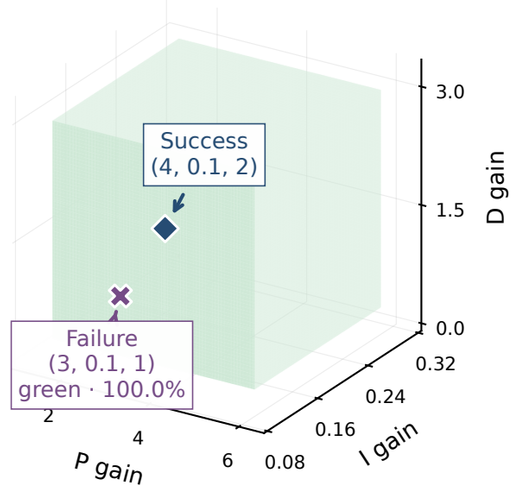
Roll / Pitch
Success: green (100%)



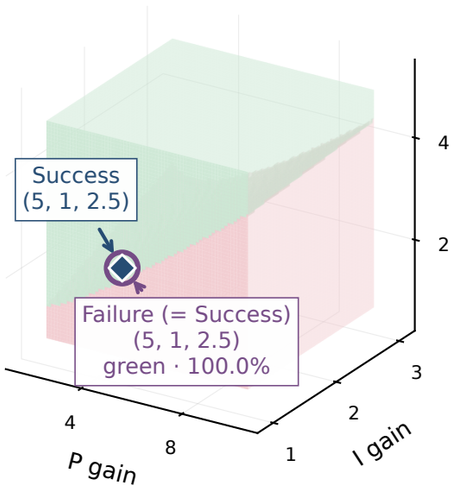
Yaw
Success: green (100%)



XY
Success: green (100%)



Z
Success: green (100%)



Green region: survival ≥ 95% Red region: survival < 95%

3D volumes are cut at each shared reference I to show both points. Purple marks identify the failure-flight gains; Z gains coincide.

Each case uses its own posterior. Other controller gains are fixed at the successful-flight values.